



Robust laser vision sensing technology for adaptive weld seam tracking in shipbuilding automation

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Abstract

This paper presents the design and experimental validation of an automated weld seam tracking system integrating a laser vision sensor with a multi-axis gantry-based welding robot. In automated welding, accurately detecting feature points of the weld groove from laser stripe images is crucial yet challenging owing to noise from spatter, fumes, and specular reflections. To address this challenge, a robust noise filtering algorithm combined with the YOLOv8-pose deep learning model is introduced. The YOLOv8-pose model provides a high inference speed and detects keypoints and their interconnected skeletal structures analogous to joints. In the proposed approach, the laser stripe generated by the laser vision sensor is considered a skeletal structure, and critical feature points necessary for seam tracking are defined as the corresponding keypoints. By specifically training the model to recognize the unique geometric characteristics of the laser stripe, the weld seam is clearly distinguished from surrounding noise, enabling precise and real-time seam tracking. Furthermore, a supporting algorithm mitigates residual noise induced by fumes or tack welds that cannot be fully filtered by YOLOv8-pose by detecting abnormal geometric variations in the laser stripe. Experimental results demonstrate that the proposed approach achieves a high keypoint detection accuracy with a mean absolute error of 0.128 mm, substantially enhancing productivity and ensuring consistent weld quality in industrial shipbuilding environments.

Keywords Laser vision sensing · Weld seam tracking · Deep learning · Welding automation · Image processing

1 Introduction

The shipbuilding industry is a complex and labor-intensive sector that involves the construction of large-scale structures. One critical stage in shipbuilding is the subassembly process, which is vital in forming the fundamental structure of a hull. During this stage, plates and components of various shapes are assembled and welded together to create ship modules. Currently, subassembly processes in hull manufacturing are performed manually, requiring a high level of worker expertise and repetitive welding tasks to ensure production efficiency. However, the repetitive nature of these tasks can lead to worker fatigue, possibly reducing productivity and quality. Furthermore, addressing labor shortages in such environments is challenging.

Automated welding robots have been extensively studied and implemented as effective solutions to the challenges of the manual approach [1–4]. These systems reduce the workload on workers, enhance productivity, and ensure consistent weld quality. Furthermore, these systems adapt to welding materials and thickness variations, making them suitable for subassembly processes. Most related research has been focused on welding robots that utilize welding carriages, articulated manipulators, or crane and gantry systems for large-scale subassembly component welding. For example, Lee et al. [5, 6] proposed a welding robot system called RRXC to address the challenges of automating inaccessible areas. RRXC integrates a six-axis modular controller, serial manipulator with three prismatic and three revolute joints, and auxiliary transport device. By establishing simulated welding paths, RRXC enables automated welding inside double-hull blocks using a carriage and articulated robot. Mun et al. [7] developed an automated welding system that incorporates a five-axis gantry system and six-axis welding robot capable of welding subassembly components within a 12 × 16 m area. By utilizing rotation-capable torches with a

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welding speed up to 80 cm/min and impact sensors to prevent robot damage, their system provides improved productivity and safety in the workplace, contributing to address labor shortages. Despite the advancements in automated welding systems, real-time weld seam tracking in subassembly automation solutions has not been properly developed.

Traditional subassembly welding relies primarily on predetermined or simulation-based path planning, which utilizes predesigned data to follow a fixed welding trajectory. However, this conventional approach fails to dynamically adapt to changes in the work environment. The lack of real-time weld seam tracking often results in errors and suboptimal responses to abnormal conditions during operations. The lack of real-time weld seam tracking can reduce the welding quality and efficiency [8, 9]. In addition, variations in the position or shape of the workpiece, caused by machining errors or welding heat deformation, can misalign the welding torch, resulting in incomplete joints or the need for rework. These problems can compromise the mechanical strength of subassembly components, ultimately influencing the process productivity and reliability. Moreover, any path misalignment requires additional adjustments, increasing the operation time and resource consumption.

To address existing challenges, recent studies have focused on integrating robotic welding systems with laser vision sensors to implement adaptive automated welding [10–12]. Silva et al. [13] proposed an adaptive automated welding system that utilizes laser vision sensors to measure weld profiles, enabling real-time path correction during gas metal arc welding (GMAW). This system enhanced the welding accuracy by compensating for irregular weld paths caused by the intense heat generated during welding. Similarly, Hou et al. [14] developed an algorithm to extract weld groove feature points with pixel-level precision using laser vision sensors and image processing techniques, achieving a tracking accuracy within 0.2 mm. Xin et al. [15] designed a lightweight vision sensor that combined a compact industrial camera and linear stripe laser projector, being suitable for robotic welding.

The technological integration of vision sensors into robotic welding systems has propelled the evolution of adaptive manufacturing in subassembly processes, leveraging the advantages of noncontact sensing and real-time data processing to enhance efficiency. However, environmental noise, such as spatter and fumes generated during welding, poses a major challenge that can compromise the reliability of vision sensor systems.

Image processing methods that rely on laser stripe patterns are highly susceptible to environmental noise. A laser vision sensor typically consists of a laser module and camera, and the weld geometry is measured by detecting feature points (e.g., corners) on the laser stripe through image processing. However, intense arc light, smoke, and spatter can notably distort the shape of the laser stripe in captured

images, hindering the extraction of a clean stripe pattern. Consequently, the image processing accuracy deteriorates, thereby reducing the seam tracking performance and weld quality. To address this problem, image processing techniques capable of isolating undistorted laser stripe patterns and minimizing noise influence must be devised. Such methods may help reduce errors in the processing pipeline and ultimately enhance the accuracy of weld seam tracking.

Reflected or scattered laser beams on metal surfaces can degrade the signal accuracy of laser sensors [16]. Furthermore, intense arc light and spatter during welding introduce noise, compromising the reliability of the vision system. The misidentification of weld seams when passing through tack welds on large hull areas can also negatively impact sensor reliability. Hence, we introduce an automated welding system that utilizes laser vision sensing to perform accurate and robust real-time weld seam tracking in ship hull welding processes. The proposed laser vision sensor is optically designed to suppress specular reflection on metal surfaces while precisely measuring weld profiles, thereby enabling high-precision weld seam tracking.

To accurately extract weld profiles, an image-based deep learning model was utilized to identify weld seam feature points. A vision processing algorithm was developed to enable reliable online detection and tracking of weld seam feature points, even under image distortion caused by spatter, fumes, and tack welds. Finally, the developed laser vision sensor was integrated into a multi-axis gantry-based automated welding system to perform large-area hull welding tasks, and the stability and reliability of the system were experimentally validated. This study was aimed to address the key technical challenges in the practical implementation of welding automation and considerably enhance the efficiency of high-precision large-scale welding processes in the shipbuilding industry.

Section 2 describes the optical design of the laser vision sensor, and Section 3 explains the image processing and deep learning algorithms for extracting feature points from laser stripe images. Section 4 discusses the configuration of the weld seam tracking system based on a laser vision sensor. The system is experimentally validated through welding tests presented in Section 5. Finally, we draw conclusions in Section 6 for completeness.

2 Laser vision sensing for automated welding system

2.1 Configuration of automated weld seam tracking system

We present a novel configuration for an automated hull welding system utilizing a laser vision sensor. The proposed

system is illustrated in Fig. 1 and meticulously designed around a large-scale multi-axis gantry system to facilitate extensive hull welding operations. To ensure precise welding position settings, the system integrates a multi-joint welding robot with the carriage and motion plate of a multi-axis gantry. A welding torch is fixed to the end effector of the welding robot, with a laser vision sensor positioned parallel to the welding direction at the front of the torch. The laser vision sensor measures the weld joint geometry ahead of the welding torch and performs seam tracking.

The laser vision sensor captures images of the weld joint geometry, which are then transmitted to a sensor controller housed in the control system panel located on the side of the gantry system. Controller-operated software manages the vision sensor, and image processing is applied to analyze the weld joint geometry and extract the weld seam position data. The extracted weld seam position data are transmitted to the motion control module of the welding robot and gantry system, where the offset from the reference weld seam position is calculated. The calculated offset is corrected in real time, enabling precise motion control of the welding robot. Notably, the system improves the accuracy of seam tracking by analyzing the coordinates of the weld seam along the lateral and depth axes, allowing for precise offset measurements relative to the reference weld seam.

2.2 Optical design and optimization for laser vision sensors

Selecting the appropriate laser for a laser vision sensor is crucial in guaranteeing the precision and dependability of

weld seam tracking. The red laser is commonly utilized in vision sensors owing to its wavelength range of 620–750 nm that offers high visibility and simplifies setup and alignment [17]. The red laser is the standard for most industrial vision sensors. However, in environments involving highly reflective materials, such as during aluminum welding, the performance of red laser can degrade owing to noise caused by reflections and other artifacts [18].

We adopt a blue laser for the laser vision sensor to overcome the challenges posed by strong reflections. Blue laser, with a short wavelength of approximately 450 nm, can minimize scattering and enhance the signal-to-noise ratio on reflective metal surfaces, such as stainless-steel surfaces widely used in shipbuilding. Blue laser is particularly effective on machined grooved surfaces that frequently show specular reflections. By minimizing noise to improve detection stability, the same laser is applicable to highly reflective materials, such as aluminum, which are prone to intense optical noise due to surface reflectivity [19]. Additionally, the blue laser can generate a narrower spot than the red laser, improving the spatial resolution for more precise weld seam extraction. Furthermore, the blue laser has higher absorption rates on reflective materials, reducing the signal loss from reflections and enabling stable detection of the weld seam geometry [20]. The laser vision sensor developed in this study leverages the advantages of blue laser to overcome various imaging challenges, including reflection, spatter, and fumes in welding environments, thereby ensuring accurate weld seam tracking.

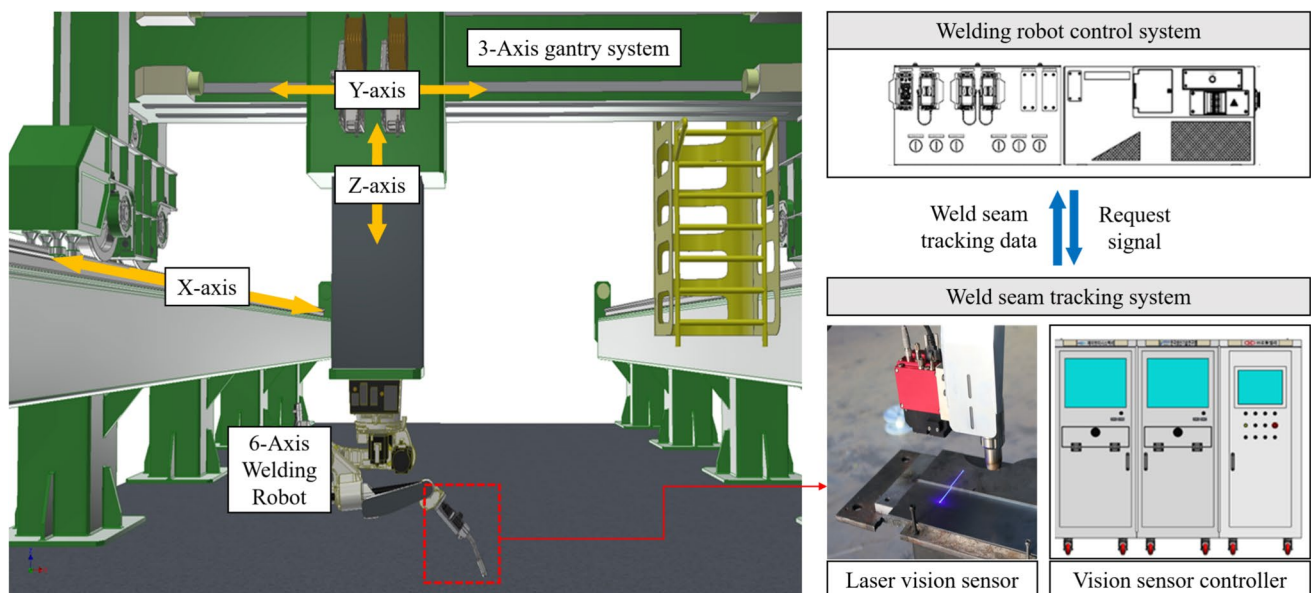


Fig. 1 Diagram of 3D multi-axis gantry-based automatic weld line tracking and high-speed welding system

Table 1 Characteristics of red and blue lasers

Characteristic	Red laser	Blue laser
Wavelength	620–750 nm	400–450 nm
Spot size	Large	Small
Noise in reflective surfaces	High	Low
Scattering	Moderate (prone to reflection)	Superior (reduced scattering)

A comparison of red and blue lasers is presented in Table 1, and their imaging performance on a machined welding grooved specimen is shown in Fig. 2. The images reveal that red laser exhibits considerable noise around the laser stripe and has a lower density of stripes compared with blue laser. In contrast, blue laser demonstrates less scattering and a denser stripe pattern, likely enhancing weld joint recognition through image processing.

The laser vision sensor utilized the UI-5140CP-M-GL Rev. 2 camera (iDS, Germany) with specifications listed in Table 2. The optical lens was the LM16JC model (KOWA, Japan) with specifications listed in Table 3. Additionally, an optical filter corresponding to the blue laser wavelength was attached to the front of the optical lens to exclusively capture the laser stripe reflected from the weld surface as image data. This configuration suppressed ambient light, allowing for the exclusive acquisition of laser stripe images, as shown in Fig. 3. The resolution of the acquired images was approximately 0.032

Table 2 Parameters of camera module

Parameter	Specification/value
Model	UI-5140CP-M-GL Rev.2 (iDS, Germany)
Exposure time	10.0 μ s
Resolution (H $\times$ V)	1280 $\times$ 1024 (1.3 magapixels)
Pixel size (horizontal)	1.315 μ m
Pixel size (vertical)	1.275 μ m
Frame rate	24 fps

Table 3 Parameters of optical lens

Parameter	Specification/value
Model	LM16JC (KOWA, Japan)
Aperture (F-stop)	F1.4–F16
Mount	C-mount
Focal length	16 mm

mm/pixel along the horizontal axis and 0.031 mm/pixel along the vertical axis in the image coordinate system.

3 Algorithm for laser vision sensing

To perform weld seam tracking, a reference point must be identified from the laser stripe image by recognizing the image features of the laser stripe. The weld seam tracking

Fig. 2 Specular reflection and scattering characteristics on images of stainless-steel surface for structured lasers with different wavelengths

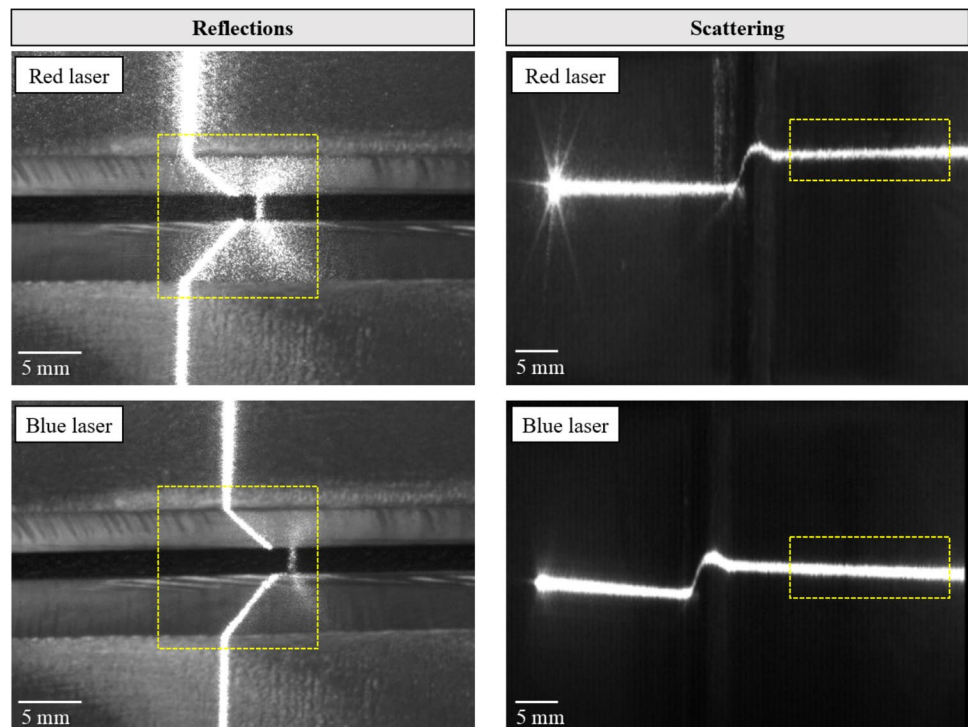
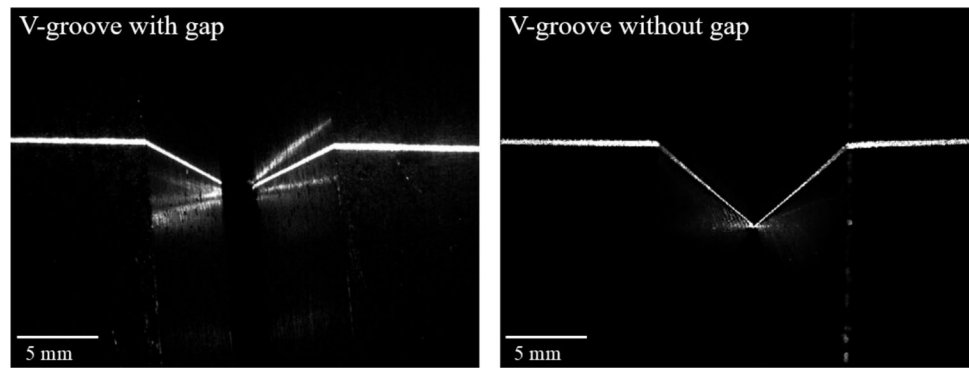


Fig. 3 Laser stripe images captured using optically optimized laser vision sensor



algorithm assesses the deviation of the welding robot from the weld centerline by detecting variations in the identified feature points and subsequently adjusting the robot position. The coordinates of the weld seam feature points are crucial for determining the robot position, facilitating path adjustments, and enhancing the tracking accuracy.

We use a deep learning image processing algorithm to accurately detect weld seam feature points from a laser stripe by identifying the edges or centerline of the weld groove. Furthermore, a noise filtering algorithm is proposed to minimize the image noise caused by specular reflection from the machined grooved surface and spatter and fumes generated during welding. This algorithm ensures strong performance under various noise and interference sources in the welding environment, ultimately enhancing the weld quality and process stability. By eliminating visual noise caused by specular reflection, noise filtering enhances the detection accuracy. Additionally, deep learning detection of groove feature points identifies three keypoints from the filtered images, enabling precise detection of the groove feature points.

3.1 Image-processing-based noise filtering

The structured laser stripe projected by the vision sensor reflects the geometry of the weld groove, as shown in Fig. 3. However, during actual welding, distortions in the laser stripe shape may occur owing to specular reflection from the metal surface and laser scattering noise. To minimize these distortions and enhance feature point detection accuracy, preprocessing must be performed to detect and minimize noise.

The steps of the image processing algorithm employed to reduce noise caused by specular reflection and laser scattering in the laser stripe image are illustrated in Fig. 4. In the original laser stripe image shown in Fig. 4a, distortions resulting from specular reflections are apparent. These noise patterns closely resemble the laser stripe, potentially leading to large errors during feature point detection. The binarized laser stripe image obtained by applying a specific threshold is shown in Fig. 4b. In the original image, the laser stripe

appears substantially brighter than the surroundings. By suppressing pixels with brightness levels below the threshold, part of the noise is eliminated. We set a fixed binary threshold of 160 to distinguish the laser stripe from the background. The application of a threshold corresponding to the brightness of the laser stripe drastically reduces fine-grained noise around the stripe. Furthermore, the background noise outside the laser stripe and its surroundings is eliminated, resulting in a laser stripe image with clear contours.

The steps of the algorithm utilized to eliminate prominent noise caused by specular reflection is shown in Fig. 4c and d. Specular reflection noise typically appears thicker than the laser stripe itself considering the stripe thickness along the vertical direction in the image coordinates. By thinning the laser stripe using its average thickness t , the algorithm can distinguish the stripe from specular reflection noise. The results of the thinning algorithm for the binarized image using average thickness t of the laser stripe are shown in Fig. 4c.

The thinning algorithm is formulated in Eqs. (1) and (2). Thinning involves scanning all columns of the binarized image and recording the rows where the sum of the pixels, $S(i, k)$, within a sliding window reaches its maximum, as defined in Eq. (1), where i denotes the column being scanned and k is the row position within column i . $S(i, k)$ is the total pixel value from row k to $k + t$ within column i . Using Eq. (2), the algorithm identifies row k with the maximum pixel sum per column, thereby locating the centerline of the laser stripe.

$$S(i, k) = \sum_{j=k}^{k+t-1} I(i, j) \quad (1)$$

$$k_{\max}(i) = \operatorname{argmax}_k S(i, k) + \frac{t}{2} \quad (2)$$

In Eq. (3), $C(i)$ denotes the set of row positions corresponding to the laser stripe centerline in each column obtained from thinning. The size of this set aligns with the number of columns in the binarized image. The thinned

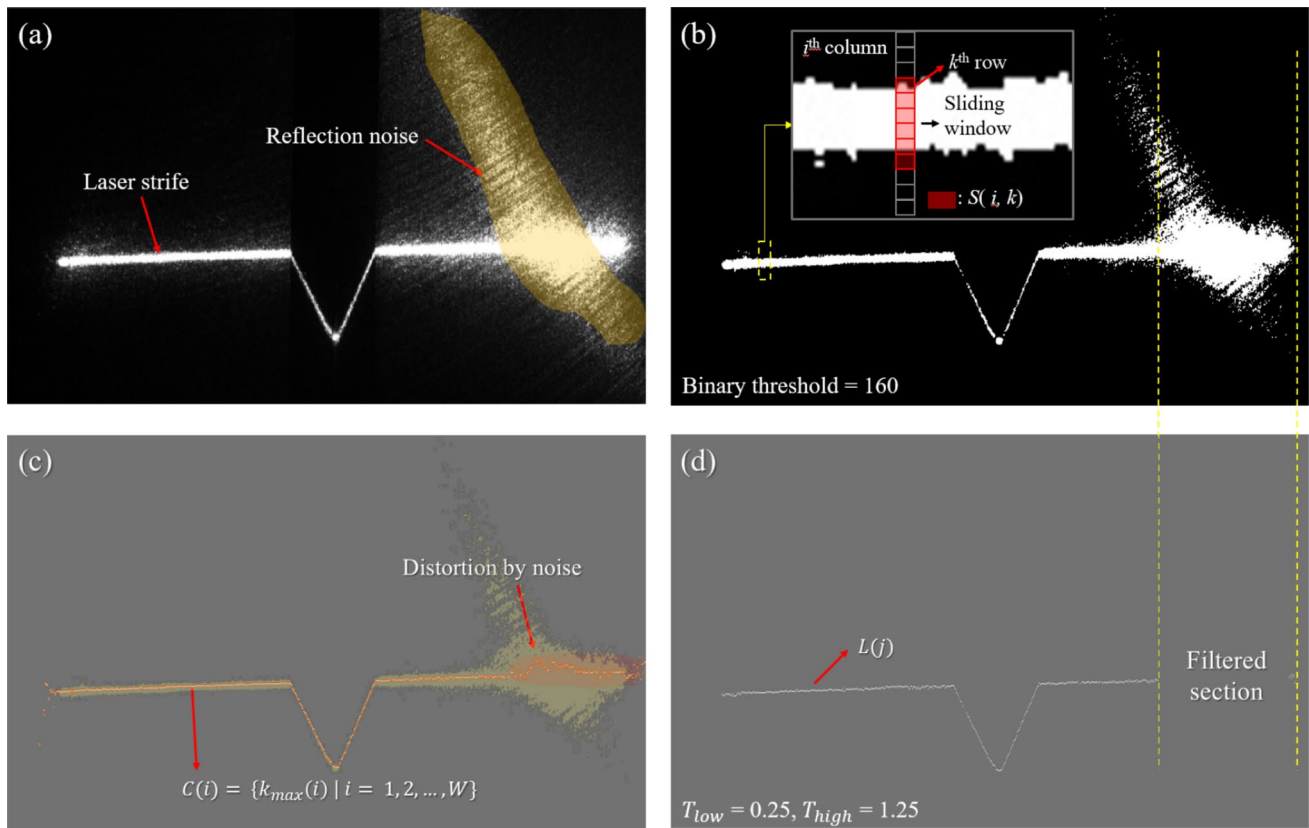


Fig. 4 Weld seam extraction and noise filtering algorithms. **a** Original laser stripe image, **b** binarized image, **c** laser stripe centerline image obtained through thinning applied to binarized image, and **d** laser stripe centerline image after noise filtering

image generated by the proposed algorithm is shown in Fig. 4c. If the algorithm detects multiple rows as centerline positions, the median across these rows is selected as the centerline row for the laser stripe.

$$C(i) = \{k_{\max}(i) | i = 0, 1, \dots, W\} \quad (3)$$

Despite applying thinning, shape distortion caused by specular reflection noise may persist. To address this problem, a novel image processing algorithm distinguishes the laser stripe to suppress noise. As shown in Fig. 5, for column $S(i_3)$ influenced by noise, such as specular reflection or spatter, the sum of the total pixel brightness across the column is substantially higher than that of the column $S(i_2)$, which contains only the laser stripe. Columns $S(i_1)$ and $S(i_2)$ with only the laser stripe consistently have similar brightness sums. Therefore, the algorithm distinguishes the laser stripe from noise based on the brightness sums. The brightness sum per column is defined using Eq. (4), where $S(i)$ is the sum of pixel brightness values for column i , h is the number of rows in the binarized image, and $I(i, j)$ indicates the brightness of the pixel at the i -th row in column i .

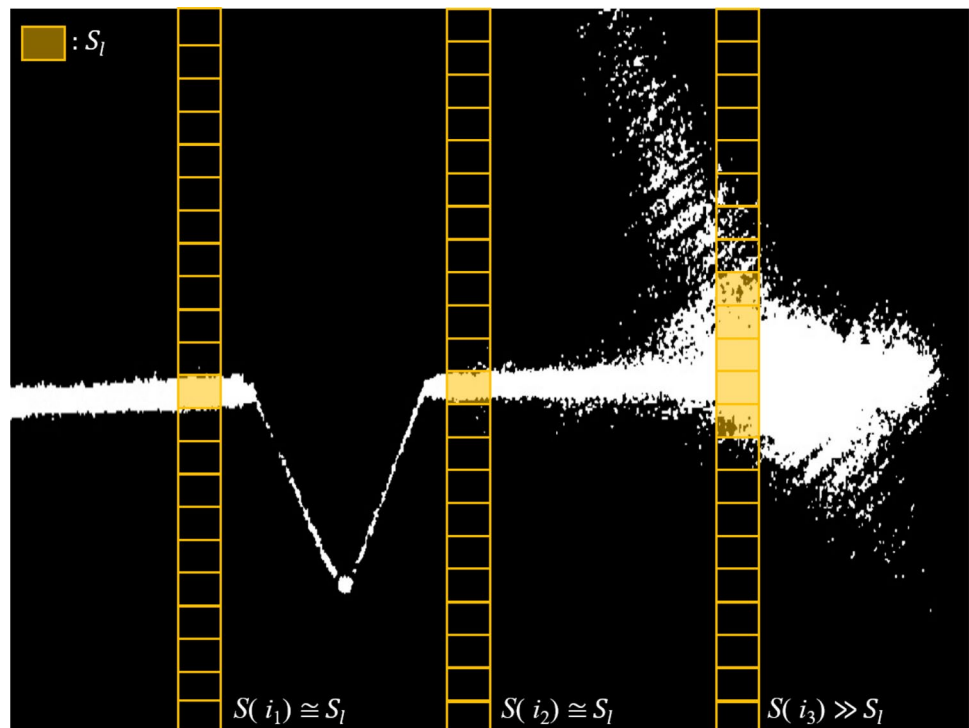
The sum of the brightness values, $S(i)$, per column calculated using Eq. (4) is compared with the standard brightness sum, S_l , to determine the presence of noise. Standard brightness sum S_l is calculated using Eq. (5), where $I_{\max}$ is the maximum brightness in the binarized image and t is the pixel thickness of the laser stripe, as defined in the thinning process. We set t to 12, representing the thickness of the laser stripe in pixels.

$$S(i) = \sum_{j=0}^{h-t+1} I(i, j) \quad (4)$$

$$S_l = I_{\max} \cdot t \quad (5)$$

In Eq. (6), if brightness sum $S(i)$ for a given column falls within a specified range relative to S_l , the column is identified as containing only the laser stripe. In contrast, if $S(i)$ falls outside this range, the column is classified as containing noise. T_{low} and T_{high} are the lower and upper thresholds, respectively. In Fig. 4d, T_{low} and T_{high} are 0.25 and 1.25, respectively, to distinguish valid laser stripe regions

Fig. 5 Difference in pixel brightness sums between laser stripe and noise columns in binary image. The noise columns show notably higher pixel brightness sums, whereas the laser stripe columns have similar pixel brightness sums



from noise. These thresholds account for minor variations in $S(i)$ owing to inherent inconsistencies in the binarized laser stripe and allow to effectively filter out noise.

Let $R(i)$ be the noise identification variable for column i , where a value of 1 indicates the presence of only the laser stripe and a value of 0 indicates detected noise. As shown in Eq. (7), masking is applied using an AND operation with thinned pixel data to eliminate unnecessary noise by excluding pixels that fail to satisfy the specified conditions.

$$R(i) = \begin{cases} 1, S_l \cdot T_{low} < S(i) < S_l \cdot T_{high} \\ 0, S_l \cdot T_{low} > S(i) \text{ or } S_l \cdot T_{high} < S(i) \end{cases} \quad (6)$$

$$L(i) = R(i) \text{AND } C(i) \quad (7)$$

The complete implementation logic of image processing, including the thinning and noise filtering procedures described in Eqs. (1)–(7), is presented as pseudocode in Algorithms 1 and 2. These algorithms are designed to extract and validate the centerline of the laser stripe in columns from binarized images.

Algorithm 1 processes each column i of binarized laser stripe image $I(i, j)$ using a sliding window of t pixels in height along the row direction. For each column,

the window with the highest cumulative pixel intensity is identified. The vertical center of this window is recorded as the estimated centerline for column i and stored in array $C(i)$. The variables used in this process include pixel sum S_2 in the current window, sum S_1 in the previous window, maximum cumulative intensity S_MAX for column i , and the corresponding row index, S_MAX_ROW , of the centerline.

Algorithm 2 performs postprocessing to validate the detected centerlines based on brightness consistency. For each column, the total pixel sum $S(i)$ is computed and compared to the reference sum S_L obtained using Eq. (5) based on the maximum pixel value $I_MAX = 255$ and stripe thickness t . Thresholds T_LOW and T_HIGH define the acceptable range for valid stripe brightness. If $S(i)$ falls within this range, centerline detection is considered valid and flagged as $R(i) = 1$. Otherwise, it is marked invalid with the value $R(i) = 0$. The validated centerlines are stored in $L(i)$, while original detection $C(i)$ is retained for comparison.

Algorithms 1 and 2 were implemented using Python and OpenCV. The parameter values were empirically determined through testing across multiple image samples. We provide the pseudocode and defined parameters to ensure the reproducibility and practical applicability of the proposed method.

Algorithm 1 Pseudocode for thinning binary array of laser strife

```

1: FOR  $i$  FROM 0 TO  $W - 1$  DO
2:   SET  $S1$  TO 0
3:   SET  $S2$  TO 0
4:   SET  $S\_MAX$  TO 0
5:   SET  $S\_MAX\_ROW$  TO 0
6:   FOR  $j$  FROM 0 TO  $H - 1 - t$  DO
7:     IF  $j \leq t - 1$  THEN
8:        $S1 = S1 + I(i, j)$ 
9:        $S\_MAX = S1$ 
10:       $S\_MAX\_ROW = j + t/2$ 
11:    ELSE
12:       $S2 = S1 - I(i, j - t)$ 
13:      IF  $S2 > SMAX$  THEN
14:         $SMAX = S2$ 
15:         $SMAX\_ROW = j + t/2$ 
16:      ENDIF
17:    ENDIF
18:  ENDFOR
19:   $C(i) = SMAX\_ROW$ 
20: ENDFOR

```

Algorithm 2 Pseudocode for noise filtering

```

1: SET  $S\_L = t \times I\_MAX$ 
2: FOR  $i$  FROM 0 TO  $W - 1$  DO
3:   SET  $R(i) = 0$ 
4:   SET  $S(i) = 0$ 
5:   FOR  $j$  FROM 0 TO  $H - 1$  DO
6:      $S(i) = S(i) + I(i, j)$ 
7:   ENDFOR
8:   IF  $S(i) > S\_L \times T\_LOW$  AND  $S(i) < S\_L \times T\_HIGH$  THEN
9:      $R(i) = 1$ 
10:  ELSE
11:     $R(i) = 0$ 
12:  ENDIF
13:   $L(i) = C(i) \times R(i)$ 
14: ENDFOR

```

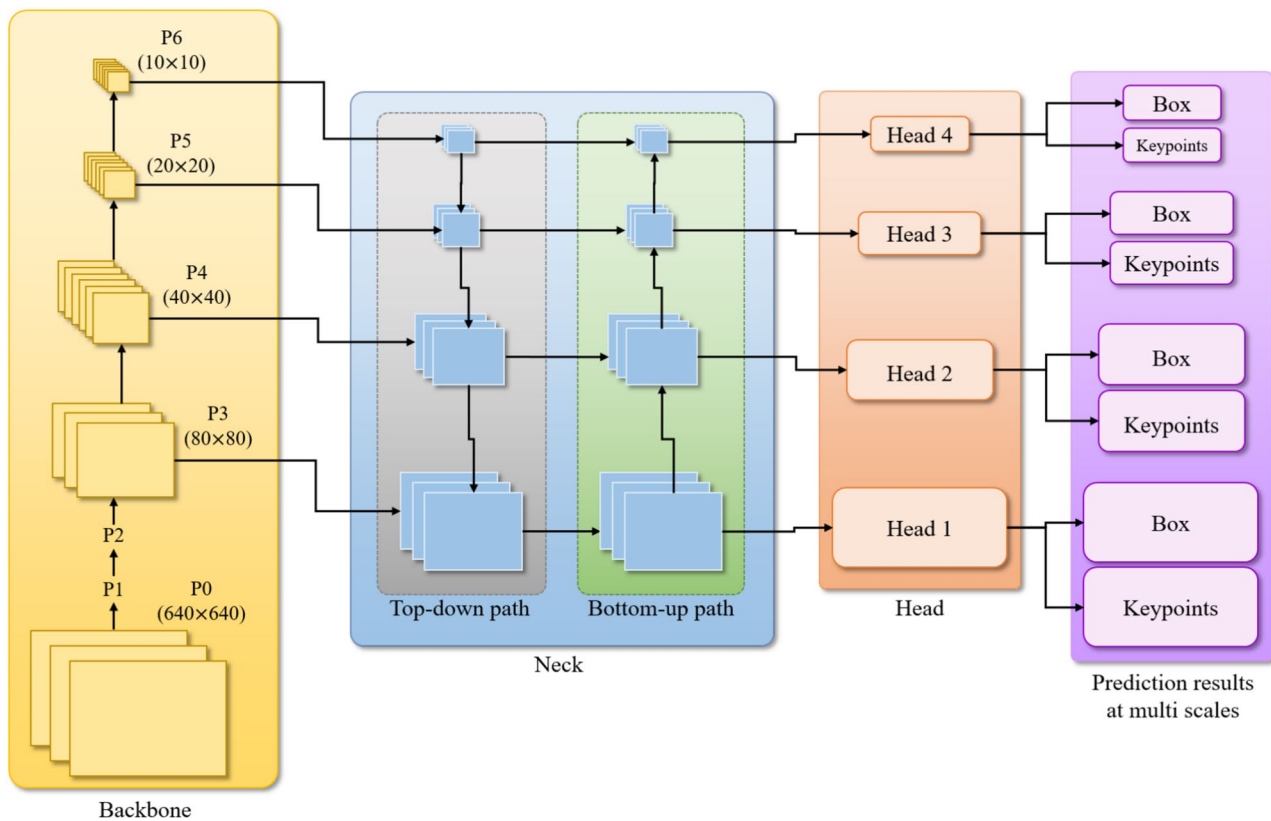


Fig. 6 Schematic of YOLOv8-pose model architecture for keypoint training, enabling detection of objects and keypoints of various sizes through multiscale feature extraction and multilevel feature fusion

3.2 Groove feature point detection based on deep learning

Conventional feature point detection algorithms based on image processing are highly sensitive to noise, which drastically reduces their accuracy. Prominent algorithms, such as Harris [21, 22] and Shi–Tomasi [23, 24] detect feature points by analyzing variations in brightness or edge contrast within the image. However, in weld groove images, the brightness variation around feature points is minimal compared with noise signals, hindering detection.

Achieving precise detection of groove feature points requires meticulous adjustment of various algorithm parameters. This iterative process is time-consuming and labor-intensive, particularly when application environments or conditions change frequently. Such inefficiency is particularly problematic in industrial settings, where real-time processing is essential. Additionally, even minor modifications to parameter settings can lead to large variations in the detection outcomes, posing challenges to performance consistency.

To address the abovementioned issues, the proposed feature point detection model is built upon the YOLOv8-pose model [25], a cutting-edge deep learning algorithm tailored

to keypoint detection. This model is suitable for extracting feature points from weld groove (V-groove) images because it learns the geometric characteristics of the laser stripe and is less affected by local brightness variations and noise components, such as spatter and arc light, than conventional image processing methods. The YOLOv8-pose model has remarkable adaptability to environmental changes through image-based learning, ensuring robust feature point detection even in noisy conditions. Notably, the YOLOv8-pose model provides fast inference, a critical attribute for real-time processing of welding image data in weld seam tracking.

As shown in Fig. 6, the YOLOv8-pose model comprises three main components: backbone, neck, and head. The backbone extracts features from the laser stripe image. It comprises multiple convolutional layers and utilizes a receptive field to extract local patterns within the image. This process leverages the pattern information around keypoints to learn and accurately determine their locations. The neck serves as an intermediary stage that processes features extracted by the backbone and delivers them to the head. The neck enhances the feature information through both bottom-up and top-down pathways, enabling effective feature learning for objects of various sizes. This process involves

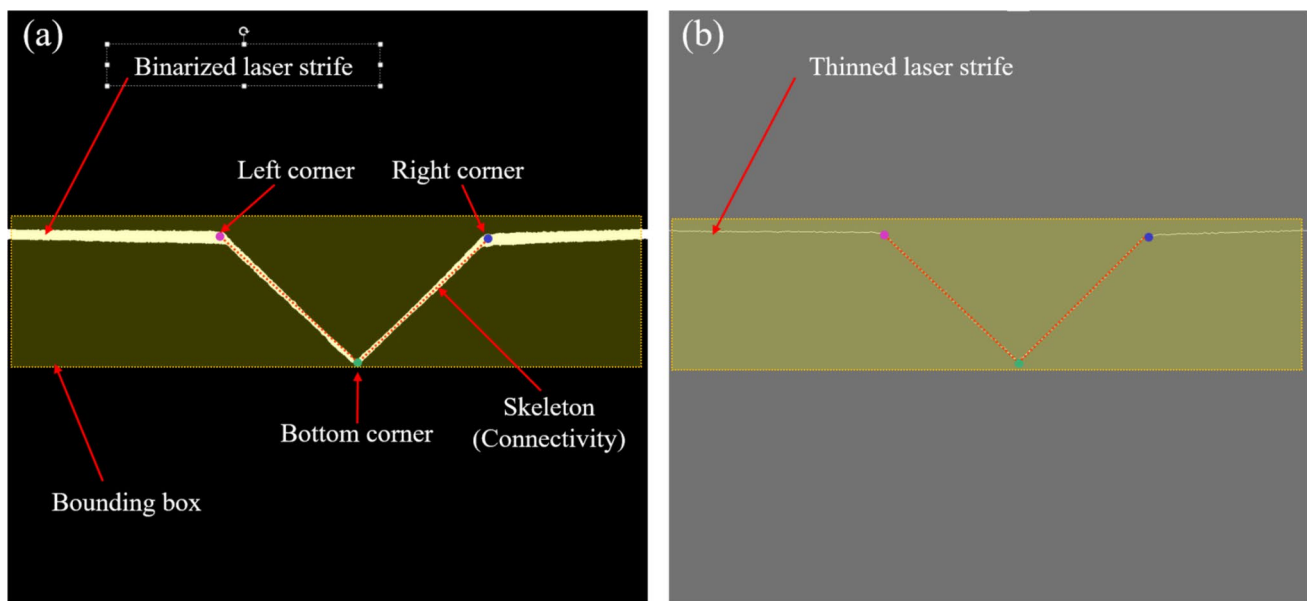


Fig. 7 Input images and annotation points for training YOLOv8-pose model. **a** Binarized and **b** thinned images with annotated feature points

integrating high-level abstract information with low-level detailed information, which is then transmitted to the head. Specifically, a path aggregation network is employed in the neck to mesh multiscale feature maps, thereby refining the features for both local and global keypoint detection. Finally, the head leverages the processed feature information from the backbone and neck to accurately detect the object bounding boxes and keypoints.

The input images for the YOLOv8-pose feature point detection model included noise-filtered thinned images and binarized images with only noise filtering applied. Both the thinned and binarized images retained feature points that represent the geometric characteristics of the laser stripe, with unnecessary noise eliminated through preprocessing. However, the visual disparities in the laser stripe representation in these images might cause issues such as loss of shape information or inadequate learning of surrounding patterns in the convolutional layers

during training of the YOLOv8-pose model. To prevent these issues, we optimized the performance of the feature point detection model by comparing the training outcomes obtained from the two types of input images.

The training data for the YOLOv8-pose model using two types of input images were structured as shown in Fig. 7. To construct the dataset, three feature points of the V-groove were annotated per preprocessed image. These feature points corresponded to the left, bottom, and right corners. The annotated feature points were positioned within the V-groove bounding box and assigned the same label as that of the bounding box to generate the training data. The dataset included binarized and thinned images, as well as the normalized coordinates of the feature points (ranging from 0 to 1) and the bounding box coordinates per image.

The deep learning model based on YOLOv8-pose was optimized by evaluating its performance with various types of input images, model weight sizes, and batch sizes as optimization parameters. The training parameters and conditions for model optimization are listed in Table 4, and the hyperparameters for all cases considered in the optimization process are listed in Table 5. The YOLOv8-pose model was trained and optimized on a workstation equipped with an NVIDIA GeForce RTX 3080 Ti graphics processing unit (GPU) with 12 GB of GDDR6X memory. The model weights were configured using the YOLOv8-pose model with small, medium, and large variants comprising approximately 11.2, 25.9, and 43.7 million trainable parameters, respectively. Excessively large model weights can result in issues such as vanishing gradients, resulting in errors in weight updates, or negligible

Table 4 Training parameters for optimizing weld feature detection model

Parameter	Value
No. training samples	438
No. validation samples	24
No. test samples	48
Optimizer	AdamW
No. epochs	1000
Weight decay	0.0005
Initial learning rate	0.01

Table 5 Training cases for optimizing feature detection model based on hyperparameters

Case	Hyperparameters		
	Input image	Model size	Batch size
1	Binarized laser stripe	Small	8
2	Binarized laser stripe	Small	16
3	Binarized laser stripe	Medium	8
4	Binarized laser stripe	Medium	16
5	Thinned laser stripe	Small	8
6	Thinned laser stripe	Small	16
7	Thinned laser stripe	Medium	8
8	Thinned laser stripe	Medium	16

training progress. On the other hand, overly small weights may hinder the model ability to effectively learn complex patterns. Therefore, determining the optimal weight was essential for model optimization.

The training was conducted using high-resolution images of 1280×1024 pixels. Considering the total number of epochs (1000), a minimum batch size of 8 was required to ensure training efficiency. However, due to the memory limitations of the GPU (RTX 3080 Ti with 12 GB in memory), the large model could only be trained with a batch size of 2–4 at this resolution, which was not suitable for stable optimization. Thus, the large model was excluded from hyperparameter tuning, and optimization was performed on the small and medium models.

Under the specified training conditions, the learning rate was set to 0.01, and the number of epochs was set to 1000. Early stopping was employed to prevent overfitting and avoid unnecessary training time. During optimization, each model was evaluated based on the error

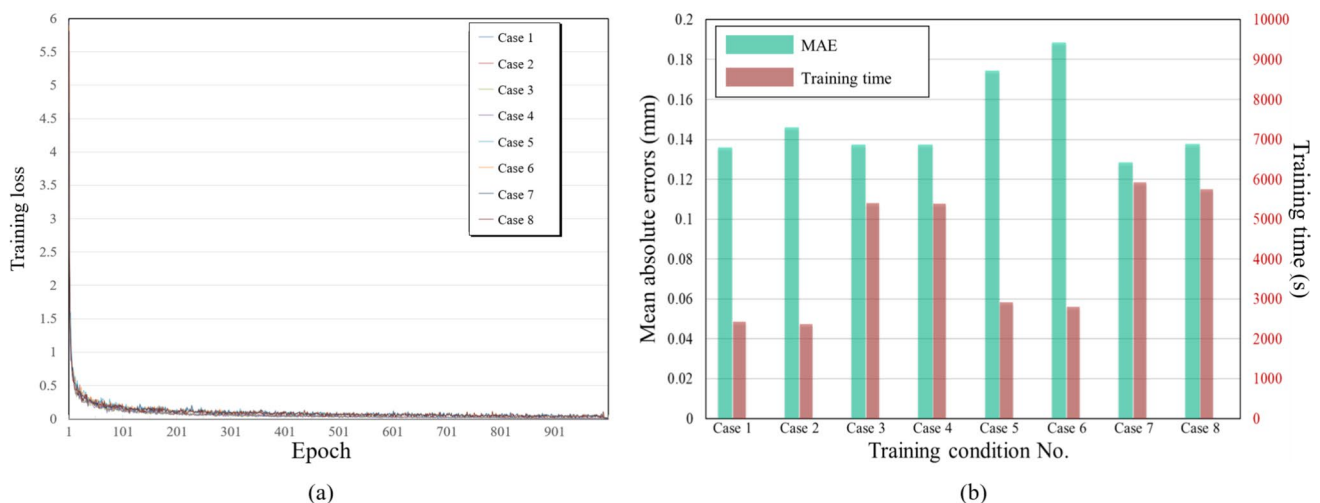
between the ground-truth feature points in the test set and those predicted by the trained model. The position error was converted from pixel units into physical distance units according to the camera resolution determined by the optical design of the vision sensor. The model performance was assessed using the mean absolute error (MAE) because the error between feature points is continuous and quantifiable (Eq. (8)).

The MAE per optimization case on the test set was calculated using Eq. (8), where $\hat{x}_i$ and $\hat{y}_i$ represent the x - and y -axes coordinates of the ground-truth keypoints of the test set, respectively, x_i and y_i represent the x - and y -axes coordinates of the keypoints detected by the trained models, and n is the number of test samples.

$$MAE = \frac{\sum_{i=0}^n \sqrt{(\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2}}{n} \quad (8)$$

The training results for all optimization cases are shown in Fig. 8, with the training loss and MAE for the test set shown in Fig. 8a and b, respectively. Among the hyperparameters, the input image type had the highest impact on the training outcomes. Models trained using binarized laser stripe images (cases 1–4) exhibited lower errors overall. Conversely, thinned laser stripe images led to higher MAE owing to substantial information loss during convolution operations. In addition to the error metrics, Fig. 8b shows the training time per case (righthand y -axis scale), allowing the simultaneous comparison of model accuracy and training time. To reduce the MAE, models of medium size were necessary for effective training.

Among the optimization cases, case 7 provided the model with the lowest MAE of 0.128 mm. Case 7 was trained using thinned laser stripe images and employed the medium-sized

**Fig. 8** **a** Training loss over epochs for each case. **b** MAE on test set (left axis) and corresponding training time (right axis)

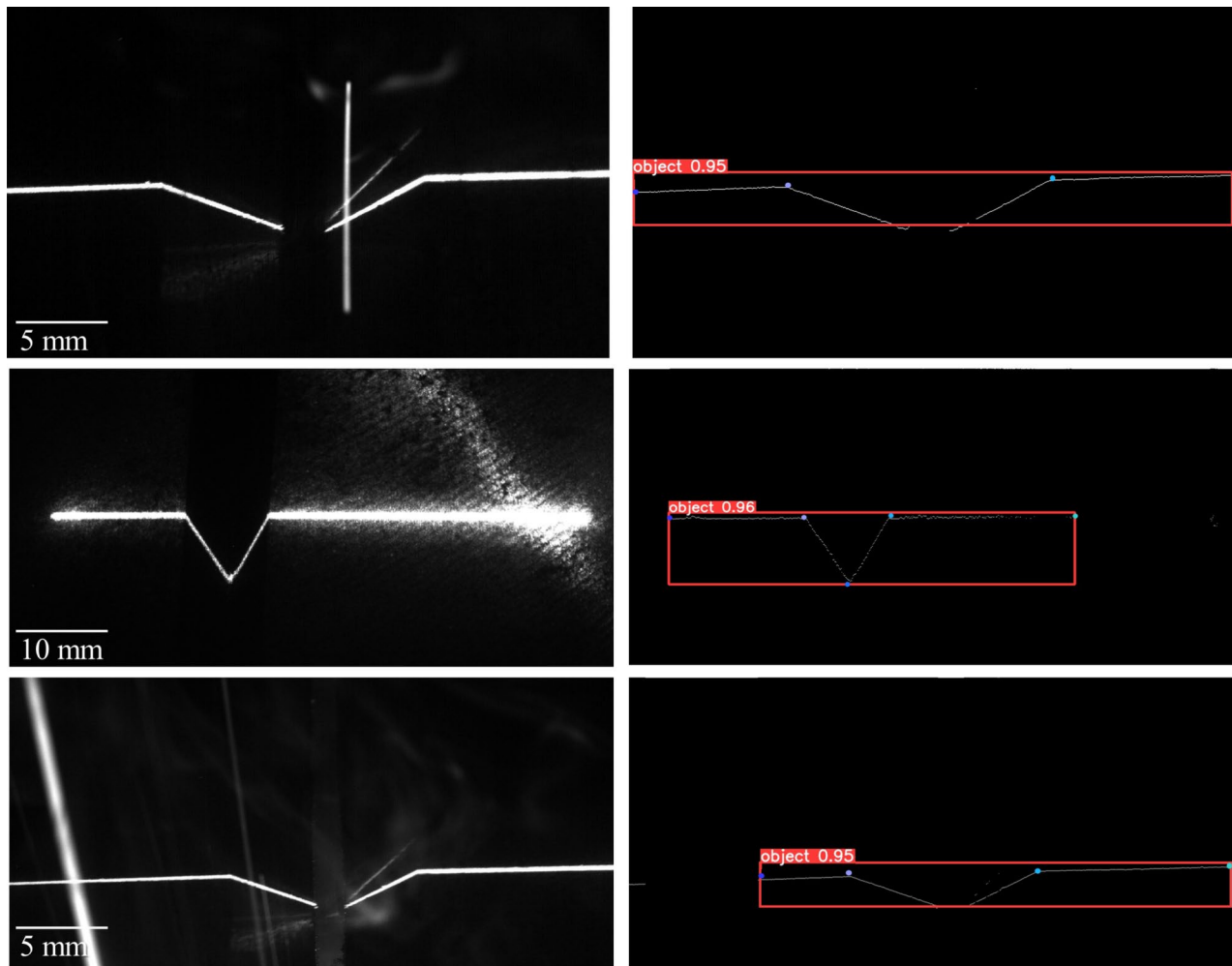


Fig. 9 Results of feature point detection in weld groove images

YOLOv8-pose model. Although thinned images generally led to higher errors, the medium-sized model in case 7 compensated such errors by preserving key structural features during training. However, as shown in Fig. 8b, the training time for case 7 was longer than that for other cases. This was attributed to the increased complexity in learning the geometric characteristics near groove corners from the thin and sparse input features of the thinned laser stripe images. In the training environment, both the small- and medium-sized models demonstrated computation times of approximately 3–5 ms, with negligible differences between them. This performance was deemed suitable for real-time weld seam tracking. Examples of feature point detection results under various welding conditions are shown in Fig. 9. The images display both the raw image and image with detected keypoints, demonstrating the algorithm successful detection of groove feature points even in noisy environments with specular reflections, spatter, and small fumes.

To evaluate the performance of the proposed keypoint detection algorithm both quantitatively and qualitatively, a comparative analysis considering related prior studies was conducted. The proposed algorithm achieved an MAE of 0.128 mm in detecting weld groove feature points from laser stripe images and an execution time of 23–28 ms, satisfying the real-time control requirement typically set below 30 ms [26]. This execution time included both conventional image preprocessing (e.g., binarization and thinning) and deep learning keypoint detection, as measured through the integrated software interface. These results suggest that the proposed algorithm can maintain fine-grained feature detection even in noisy environments.

Hou et al. [14] combined an improved Steger-based feature extraction method with RANSAC filtering, achieving a maximum tracking error of 0.19 mm along the x -axis and 0.10 mm along the z -axis under V-groove welding conditions. While this approach yielded favorable results in

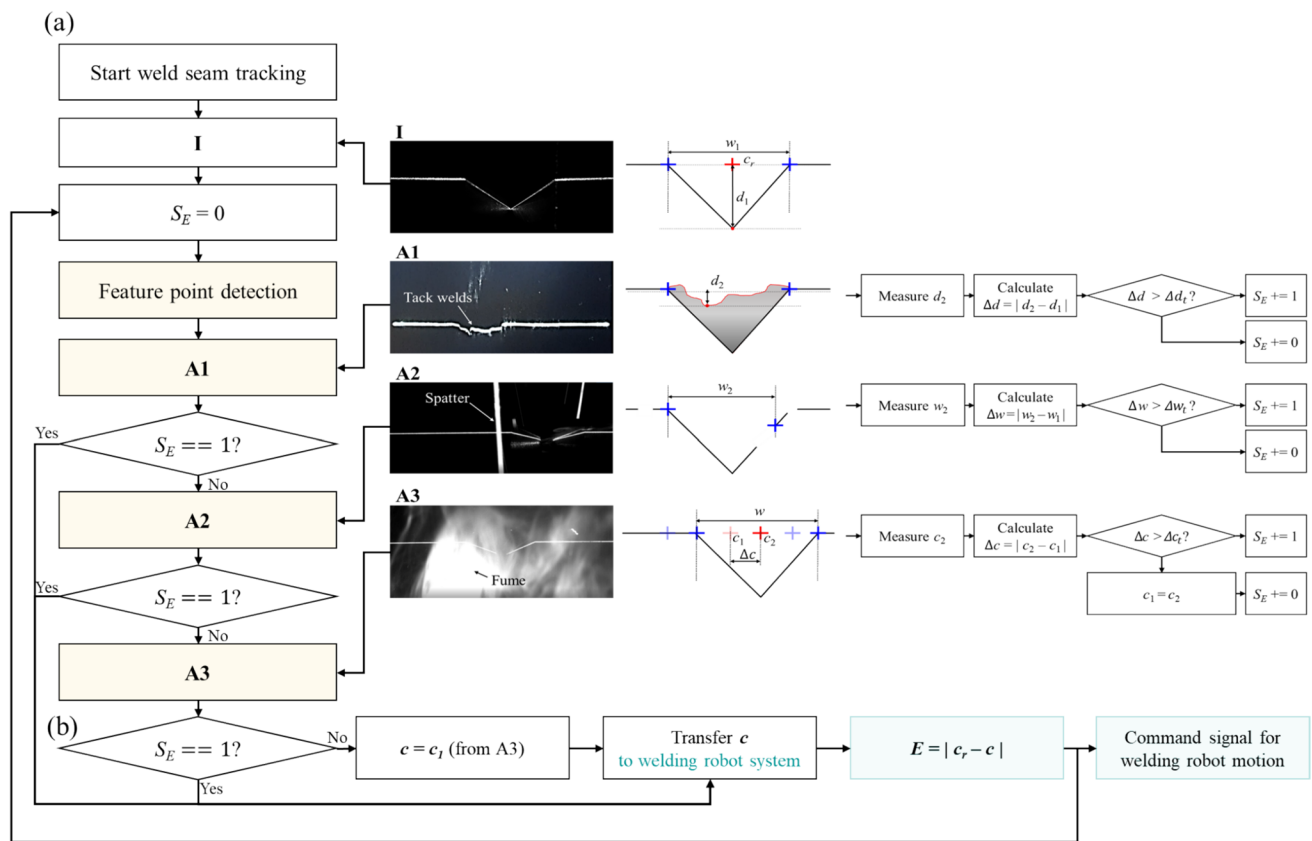


Fig. 10 Flowchart of weld seam tracking algorithm. **(a)** Noise filtering algorithm based on detection of laser stripe distortions. **(b)** Transmission algorithm for updated weld seam center point from vision sensor controller to welding robot

specific environments, its reliance on manually designed heuristic rules may limit its consistency in more complex or noisy environments.

Koçak and Saygın [27] implemented a real-time weld seam tracking system on a Jetson Nano platform using HSV color filtering and morphological image processing. Their system achieved an average MAE of 0.23 mm. Unlike our proposal, which employed blue laser, their system utilized red laser, making it more susceptible to specular reflection on metallic surfaces and potentially introducing visual noise under certain conditions. In contrast, our algorithm adopted a blue laser with a shorter wavelength, which was experimentally verified to reduce reflection and suppress noise, resulting in stable feature detection in environments containing spatter, fumes, and reflective surfaces.

Zhang et al. [28] proposed a seam-tracking algorithm designed for complex 3D curved surfaces. Their method was evaluated on both 2D symmetric and 3D curved weld lines, achieving MAEs of 0.296 and 0.292 mm, respectively. Their study focused on the applicability to geometrically complex trajectories, focusing on tracking stability and overall path planning but neglecting real-time responsiveness.

While the abovementioned studies presented different approaches optimized for specific conditions and objectives, the proposed algorithm seeks to balance real-time performance, detection accuracy, and robustness to noise. In particular, data-driven feature detection and post-processing filtering were intended to enhance adaptability to diverse and dynamic welding environments.

3.3 Weld seam tracking

Weld seam tracking involves the utilization of vision sensors during welding to automatically identify the workpiece groove and adjust the welding torch or robot position in real time. However, factors such as noise, geometric distortion, and the influence of tack welds in the welding environment can drastically reduce the accuracy of weld seam tracking. The objective of weld seam tracking extends beyond merely following the welding path, as it encompasses real-time groove position detection and adaptive path corrections to prevent deviations from the weld seam. Specifically, the goal is to minimize errors resulting from residual noise or geometric distortions in the laser stripe that may occur when

passing over tack welds, which cannot be fully addressed by the groove detection algorithm. As a result, a more precise weld seam tracking can be achieved. Accordingly, we propose an integrated weld seam tracking system interfaced with a welding robot control system.

The laser vision sensor and weld feature point detection model successfully identify keypoints. However, obstacles during weld seam tracking often lead to defects in recognizing the weld geometry. In particular, factors such as spatter, fume, and tack welds generated during welding introduce distortions in the laser stripe, substantially reducing the accuracy of feature point detection. Spatter and fume have a widespread impact on the entire image, hindering their suppression using noise filtering algorithms alone. Geometric distortions not only compromise the accuracy of weld geometry detection but also undermine the reliability of weld seam tracking. To address these challenges, a novel algorithm is introduced to eliminate noise induced by such obstacles during weld seam tracking.

The proposed algorithm to address weld seam detection errors caused by noise during weld seam tracking is described in Fig. 10a. First, feature points are detected on an undistorted laser stripe at the starting point of weld seam tracking, storing the geometric information of the weld groove (block I). The stored geometric information includes two key parameters: vertical distance d_1 between the surface boundary lines formed by the groove endpoints and lowest point of the laser stripe, and the width w_1 between the groove endpoints. Both d_1 and w_1 are fixed throughout a weld seam tracking cycle and utilized in detecting variations in geometric information caused by noise.

In the weld seam tracking algorithm, block A1 represents the algorithm developed to filter the laser stripe distortions that occur when passing over the tack welds. Tack welds, unlike unwelded sections of the seam, exhibit irregular and variable shapes, leading to fluctuations in the positions of the groove endpoints. These fluctuations can result in unstable torch movements and potential welding defects. Tack weld detection identifies these distortions by monitoring changes in the vertical distance between the detected weld groove and the lowest point of the laser stripe. During weld seam tracking, the vertical distance d_2 between the boundary formed by the feature points and the lowest point of the laser stripe is measured continuously. Difference $\Delta d = |d_2 - d_1|$ is determined, where d_1 is the vertical distance measured at the initial point of weld seam tracking. If Δd exceeds threshold Δd_t , the algorithm identifies geometric distortion caused by a tack weld at that location. Otherwise, if Δd remains below Δd_t , normal geometric detection is considered.

Block A2 is intended to address geometric distortions caused by widespread obstacles, such as spatters and fumes.

Spatter-induced distortions occur owing to the brightness of the spatter being similar to that of the laser stripe during weld seam tracking. On the other hand, fumes can create bright objects when reflecting high-intensity arc light, generating considerable noise. These bright fume objects are widespread throughout the image, often causing false detections of feature points and altering their relative positions. The algorithm continuously measures the width w_2 between two feature points during weld seam tracking and calculates difference $\Delta w = |w_2 - w_1|$, where w_1 is the width measured at the initial point of weld seam tracking. If Δw exceeds a predefined threshold Δw_t , the algorithm identifies the presence of geometric distortion caused by spatter or fume. Otherwise, if Δw remains below Δw_t , normal geometric detection is considered.

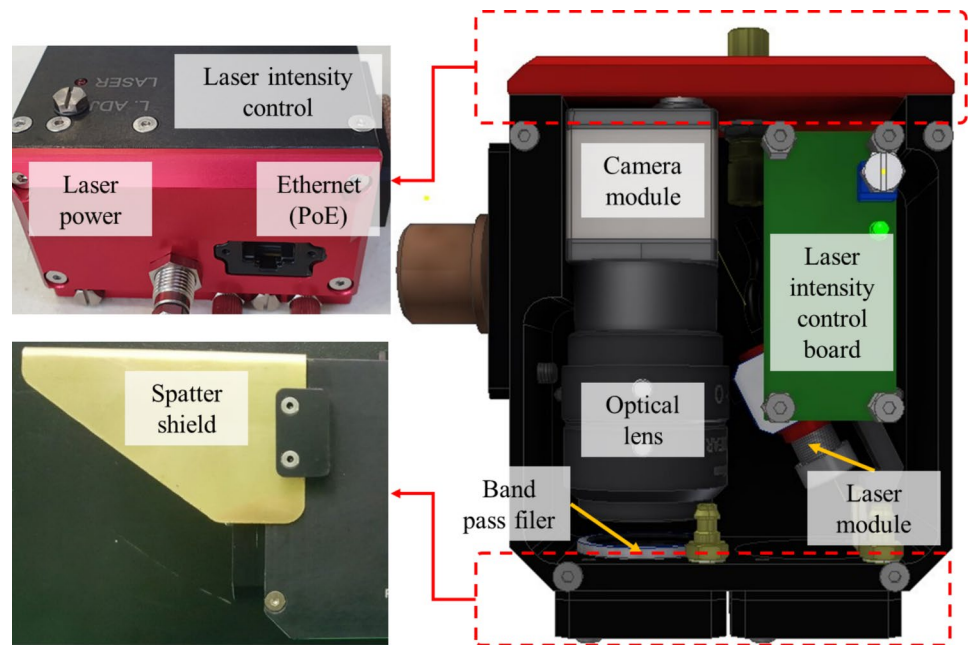
Block A3 is intended to address noise caused by the false detection of feature points resulting from spatter and fume. Block A3 complements A2 and differs from A1 and A2 because it detects weld seam tracking errors through real-time updates of geometric information. When the positions of feature points shift owing to geometric distortion caused by spatter or fume, but the width variation is negligible, detection by block A2 may be ineffective. To address such instances, block A3 measures the changes in the midpoint positions of the feature points to identify detection errors.

During weld seam tracking, block A3 continuously measures midpoint c of the two feature points. For each tracking cycle, it calculates the current midpoint c_2 and previous midpoint c_1 , updates c_1 to c_2 and simultaneously calculates the difference $\Delta c = |c_2 - c_1|$. Block A3 can identify feature point detection errors from instantaneous changes in the midpoint position caused by spatter or fume. Under normal weld seam tracking conditions, changes in the midpoint of the feature point are typically minimal. Therefore, if Δc exceeds threshold Δc_t , the algorithm determines the occurrence of geometric distortion caused by spatter or fume.

The weld seam tracking algorithm, which incorporates the three aforementioned anomaly detection algorithms, is described in Fig. 10b. This algorithm operates in an environment in which the vision sensor and welding robot control systems are connected through wired communication. The vision sensor detects weld seam coordinate information and transmits it to the welding robot control system to enable the precise control of robot motion. At the initial execution of weld seam tracking, the midpoint coordinates stored in block I are designated as reference point c_r . The welding robot calculates correction value E by measuring the deviation between the reference point c_r and current midpoint c_2 provided by the vision sensor.

The three weld seam tracking anomaly detection algorithms are executed in real time at regular intervals

Fig. 11 Design and fabrication of compact laser vision sensor module for weld seam tracking



during weld seam tracking. Each algorithm increments the anomaly signal S_E after detecting an anomaly. If the total S_E exceeds 1, indicating that at least one algorithm has detected an anomaly, the system classifies weld seam tracking as abnormal. In such cases, the update of mid-point coordinates for transmission to the welding robot is suspended, and the previously transmitted midpoint coordinates are resent. This approach helps mitigate excessive position corrections caused by erroneous weld center point detection.

4 Configuration of weld seam tracking system

4.1 Laser vision sensor

The hardware design and fabrication of the laser vision sensor are shown in Fig. 11. The optical components, camera module, and structured laser module selected in Sect. 2 were arranged based on optical optimization to ensure the best relative positions. To safeguard the internal modules from

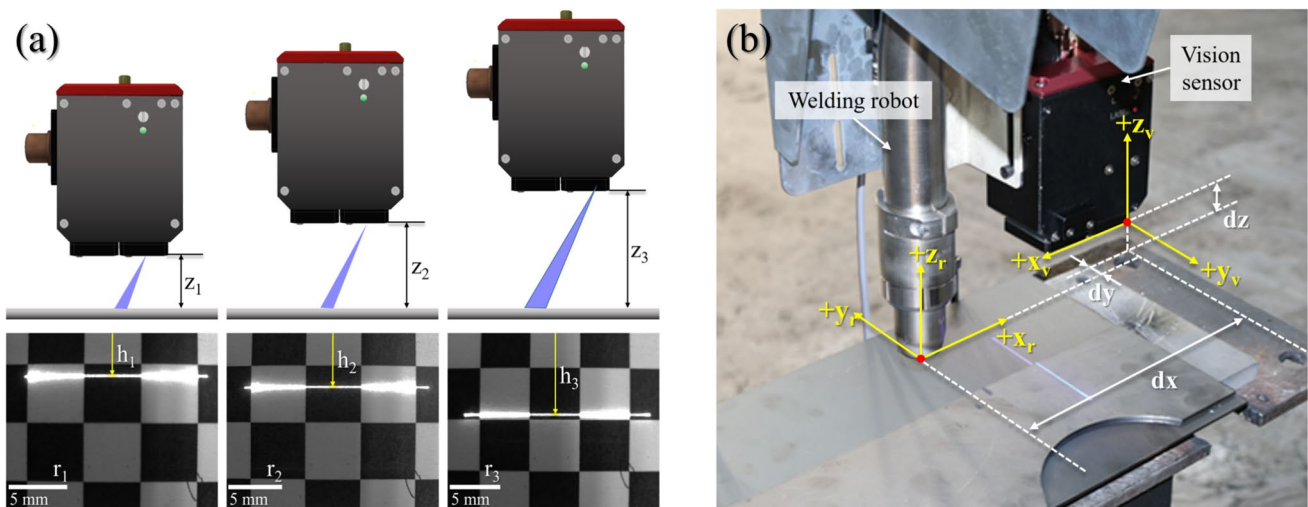


Fig. 12 Calibration for converting image coordinates into 3D coordinates. **a** Generation of calibration matrix for transforming from image coordinate system (2D) to vision sensor coordinate system (3D). **b**

Generation of 3D coordinate transformation matrix between laser vision sensor and welding robot

potential damage caused by intense arc light, fumes, and spatter during welding, an external casing was meticulously designed and constructed.

The front section of the laser vision sensor was equipped with a camera module positioned perpendicular to the surface and a laser module positioned at an angle θ for the laser beam to be captured by the camera. The structured laser module was developed to enable output adjustment through a control board.

In the rear section of the laser vision sensor, provisions were made to connect a power cable for the laser and Ethernet cable for receiving image data from the camera module. The camera module was powered through power over Ethernet, reducing cable clutter in the complex welding environment and minimizing external noise interference. Furthermore, a spatter shield was incorporated into the design and attached to the front section of the sensor, interfacing with the welding robot end effector. This shield mitigated the effects of high-intensity arc light on the camera module, reduced noise from spatter, and protected the vision sensor from damage.

As shown in Fig. 12a, simple calibration was performed to measure the 3D geometric information of the actual weld using a configured laser vision sensor. During weld seam tracking, the laser vision sensor measured coordinate variations in the lateral (y -axis) and vertical (z -axis) directions. Notably, the coordinate variation in the welding progression direction (x -axis) was not factored into tracking.

As the distance between the vision sensor and weld surface increased, the image coordinates of the laser stripe shifted, thereby establishing the correlation between the laser stripe image coordinates and 3D height (z -axis). Furthermore, in the y -axis direction, the field of view of the vision sensor changed with respect to the z -axis, resulting in variations in the resolution of the image coordinates. Consequently, a relation between 3D height z and resolution r of the image coordinates was established.

The laser stripe image was projected onto a checkerboard comprising numerous square cells, each with a side of 5 mm. The height of the vision sensor impacted both the image coordinates of the laser stripe position and number of pixels representing a single checkerboard cell. These relations were approximated using linear regression. Equation (9) describes position h of the laser stripe in the image coordinates as a linear equation, incorporating the coefficient α_z and constant β_z , both determined through linear regression. The x -axis resolution of the laser vision sensor as a function of h is calculated using Eq. (10), where α_r and β_r represent the coefficients of a linear equation derived through linear regression. The unit of r , as calculated from Eq. (10), is millimeters per pixel.

$$z(h) = \alpha_z h + \beta_z [\text{mm}] \quad (9)$$

$$r(h) = \alpha_r h + \beta_r [\text{mm/pix}] \quad (10)$$

The 3D calibration coordinate system derived from Eqs. (9) and (10) is shown in Fig. 12b. The laser vision sensor was mounted on the end effector of the welding robot for practical application. To ensure precise transmission of coordinates to the welding robot, calibration was necessary to establish the coordinate transformation between the 3D coordinate system of the laser vision sensor and that of the welding robot.

The calibration matrix between the two 3D coordinate systems is defined using Eqs. (11) and (12), where $\mathbf{R}$ represents the rotation matrix that describes the rotational alignment of the axes between the two coordinate systems relative to the origin. The coordinate systems of the laser vision sensor and welding robot are configured with a 180° rotation about the z -axis. Accordingly, the rotation matrix is implemented as shown in Eq. (11), ensuring consistency solely in the z -axis coordinate values between the systems.

Translation matrix $\mathbf{T}$ represents the translational displacement between the origins of the two coordinate systems. The reference coordinate system of the welding robot was defined at the endpoint of the wire extending from the welding robot, whereas the origin of the coordinate system of the laser vision sensor was set at the front edge of the vision sensor. The distances between these reference points in the x -, y - and z -axes, dx , dy , and dz , respectively, were measured and incorporated into the translation matrix. Transformation matrix $\mathbf{M}$ represents the homogeneous transformation between the two coordinate systems. It is constructed as a 4×4 matrix that combines the rotation and translation matrices.

$$\mathbf{M} = \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 & dx \\ 0 & -1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (11)$$

$$\mathbf{R}(180^\circ) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{T} = \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} \quad (12)$$

4.2 Integrated weld seam tracking system

The weld seam tracking system based on the laser vision sensor was integrated into a multi-axis automated welding system comprising a welding robot and gantry, as shown in Fig. 13. The laser vision sensor was attached to the end effector of the welding robot at approximately 100 mm to

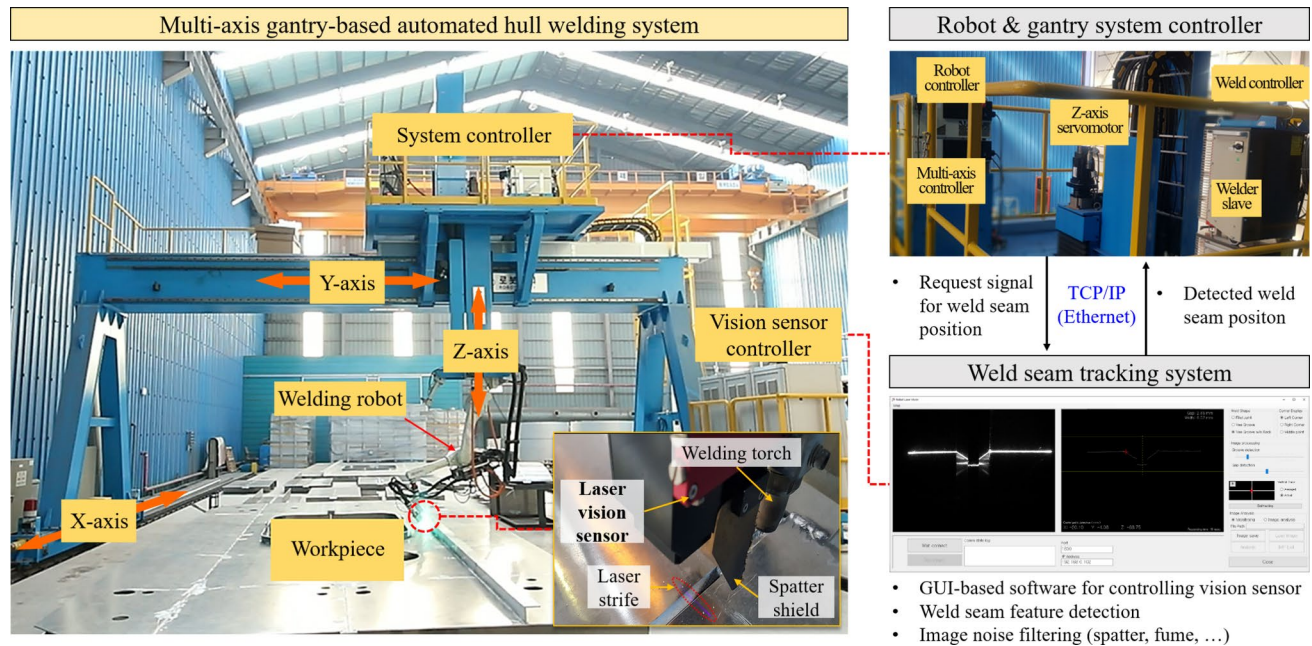


Fig. 13 Integrated hull welding system combining weld seam tracking system based on laser vision sensor and gantry-based automatic welding system

minimize the impact of intense arc light. A computer-based controller for the laser vision sensor was situated on the gantry side to process the data collected by the sensor.

The laser vision sensor controller featured a graphical user interface (GUI). The GUI allowed to analyze image data and identify groove feature points using the optimized feature detection model. The coordinates of these feature points were expressed in the 3D coordinate system of the welding robot through a calibration matrix. Subsequently, the transformed coordinates were sent to the welding robot and gantry control system through an Ethernet connection. The GUI displayed both the original images captured by the laser vision sensor and images with detected feature points. Furthermore, the GUI included a control panel that allowed users to finetune the noise filtering parameters. For data transmission and reception, the laser vision sensor and welding robot control systems utilized TCP/IP communication.

The welding robot control system operated as the server, whereas the laser vision sensor control system functioned as the client. During weld seam tracking, the server (welding robot control system) periodically requested the transmission of the weld center point coordinates. In response, the client (laser vision sensor control system) received the request signal and transmitted the updated weld center point coordinates to the server. The transmitted coordinates included only the lateral (x-axis) and vertical (z-axis) directions.

The welding robot was mounted on the y-axis frame of the three-axis gantry system to facilitate movement along the welding progression direction (x-axis). This allowed

Table 6 Specifications of multi-axis gantry-based welding system

Parameter	Specification	
Welding system	Welding robot	RA10N, E01 (Kawasaki)
	Power source	Duty cycle 60%, 420 A, GMAW, pulse
	Wire feeder	70 W, 2.5–25 m/min, ϕ 0.8–1.6 mm
	Welding torch	Torch neck 15°, TCP 50 mm
x-axis	Traveling method	Rack, pinion, and wheel
	Traveling speed	Max. 30 m/min
	Traveling stroke	25,000 mm
	Servo motor	Kawasaki, 3.1 kW $\times$ 3000 rpm
y-axis	Traveling method	Rack, pinion, and linear motion guide
	Traveling speed	Max. 30 m/min
	Traveling stroke	4550 mm
	Servo motor	Kawasaki, 3.1 kW $\times$ 3000 rpm
z-axis	Traveling method	Ball screw and linear motion guide
	Traveling speed	Max. 6 m/min
	Traveling stroke	1000 mm
	Servo motor	Kawasaki, 3.1 kW $\times$ 3000 rpm

compensating for the limited linear travel range of the welding robot, particularly when working on large and length workpieces. The welding robot control system was installed above the y-axis to prevent cable twisting or disconnection. Fine adjustments for lateral and vertical positioning

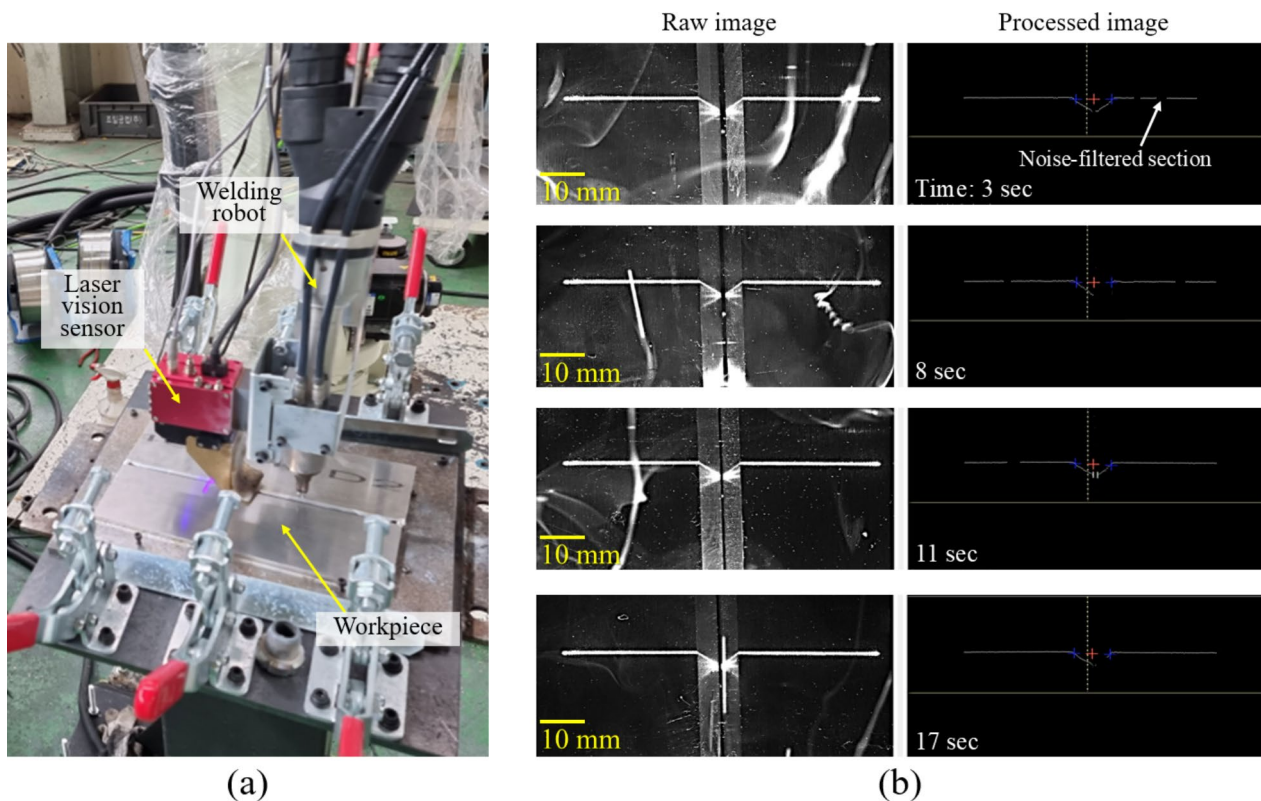


Fig. 14 **a** Experimental setup for validating performance of weld seam tracking algorithm. **b** Detection results of weld feature points obtained through weld seam tracking

Table 7 Welding and communication parameters

Parameter	Value
Workpiece	(material) 450 × 150 × 6 mm 60°
Groove type	Butt weld
Welding type	GMAW
Current	190 A
Voltage	21.4 V
Speed	90 cm/min
Wire feed	5.5 m/min
Sampling rate	20 Hz

were performed at the welding robot end. The workpiece was positioned in the subassembly area located beneath the gantry system for welding.

This configuration seamlessly integrated the weld seam tracking system based on the laser vision sensor with a multi-axis gantry-based welding system. The detailed specifications of the gantry and welding robot systems are listed in Table 6.

5 Experiments for performance validation

5.1 Validation of weld seam tracking algorithm

We conducted a welding experiment, as shown in Fig. 14a, to assess the performance of weld feature point detection and noise filtering, as well as to validate the functionality of each system component. The experiment utilized a tandem dual-wire welding method, known for generating more spatter than conventional GMAW and thus allowing for a more thorough evaluation of the noise filtering performance. The welding conditions and communication settings for transmitting and receiving the weld center point coordinates are listed in Table 7. The original laser stripe image captured during welding, detected feature points, and noise filtering results are shown in Fig. 14b. Notably, the specimen utilized in this experiment contained a gap, resulting in the bottom corner feature point being excluded from consideration.

The analysis focused on the left and right corners, which are crucial for accurate weld seam tracking. These two corners, located at the groove edges, serve as key reference points for lateral positioning. In contrast, the bottom

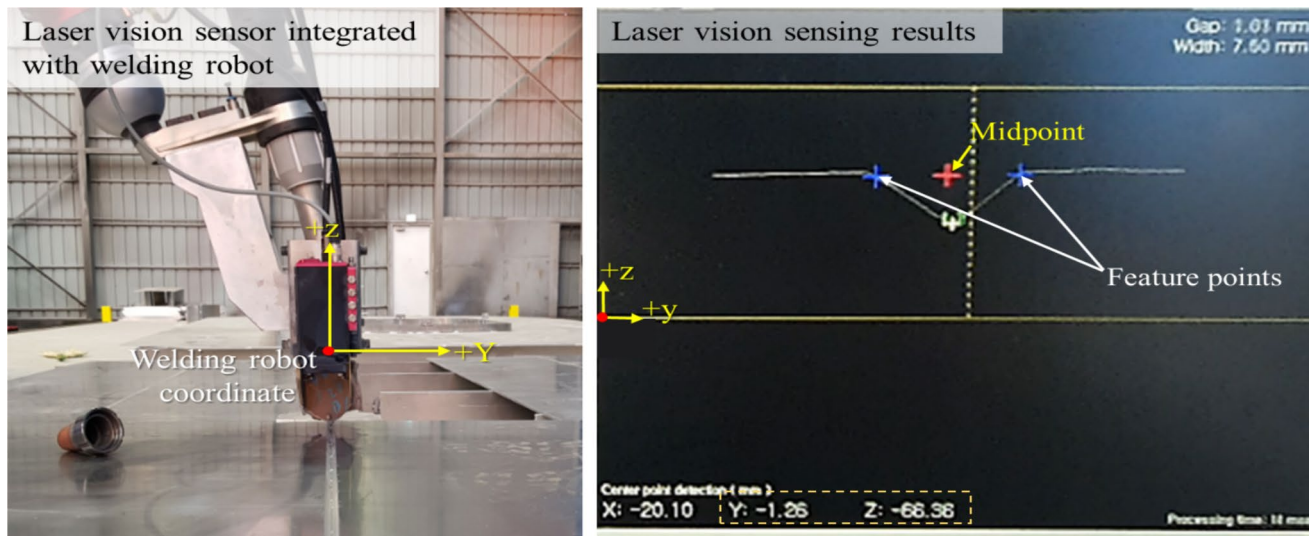


Fig. 15 Experimental setup for measuring position errors between welding robot and laser vision sensor based on displacement. The position measurement using the vision sensor was performed through the GUI for laser vision sensor control

Table 8 Transfer errors (in millimeters) between welding robot and laser vision sensor

	y: +2 (− 0.12)	y: −2 (− 4.12)	Reference point	z: +2 (− 61.86)	z: −2 (− 65.86)
y coordinate	− 0.20	− 4.01	− 2.12	− 2.17	− 1.89
y-axis error	0.08	0.11	−	0.05	0.22
z coordinate	− 63.93	− 64.02	− 63.86	− 61.82	− 66.01
z-axis error	0.07	0.16	−	0.04	0.15

corner, located on the machined grooved surface, is more susceptible to signal loss caused by strong specular reflections, particularly due to the narrow-angle intersection of two metallic surfaces. Furthermore, in cases where a gap exists between the groove faces, accurate detection of the bottom corner becomes difficult. Therefore, we limited the evaluation to the left and right corners to ensure reliable feature extraction.

Throughout welding, the presence of spatter and fume created challenges by interfering with the entire laser stripe. The spatter had a similar brightness to the laser stripe, whereas the fume reflected high-intensity arc light, resulting in comparable brightness. To address this issue, noise filtering was applied to eliminate the regions impacted by spatter and fume, leading to a partially discontinuous laser stripe.

The detected feature points accurately identified the groove endpoints and were consistently maintained throughout welding without false detections using the weld seam tracking algorithm. The proposed weld feature point detection and noise filtering algorithms reliably extracted groove feature points and weld center points under real welding conditions. This advancement can contribute to the development of high-performance weld seam tracking.

Table 9 Welding and communication parameters

Parameter	Value
Workpiece 1	SUS304 1000 × 2000 × 6 mm 60°
Workpiece 2	SUS304 852 (R-1505) × 6 mm 60°
Groove type	Butt weld
Welding type	GMAW
Current	190 A
Voltage	21.4 V
Speed	90 cm/min
Wire feed 1 (workpiece 1)	5.5 m/min
Wire feed 2 (workpiece 2)	7.0 m/min

5.2 Integrated weld seam tracking system

Welding tests were conducted to assess the weld seam tracking and welding performance of the integrated system. The measured errors between the coordinate systems of the welding robot and the laser vision sensor during the movement

of the welding torch are shown in Fig. 15. The weld center point coordinates were determined using the feature point coordinates detected by the YOLOv8-pose model. The errors in the measured displacements along the y - and z -axes are listed in Table 8. The welding torch demonstrated consistent movement of ± 2 mm along the y - and z -axes relative to the reference point, aligning with the movement of the welding robot.

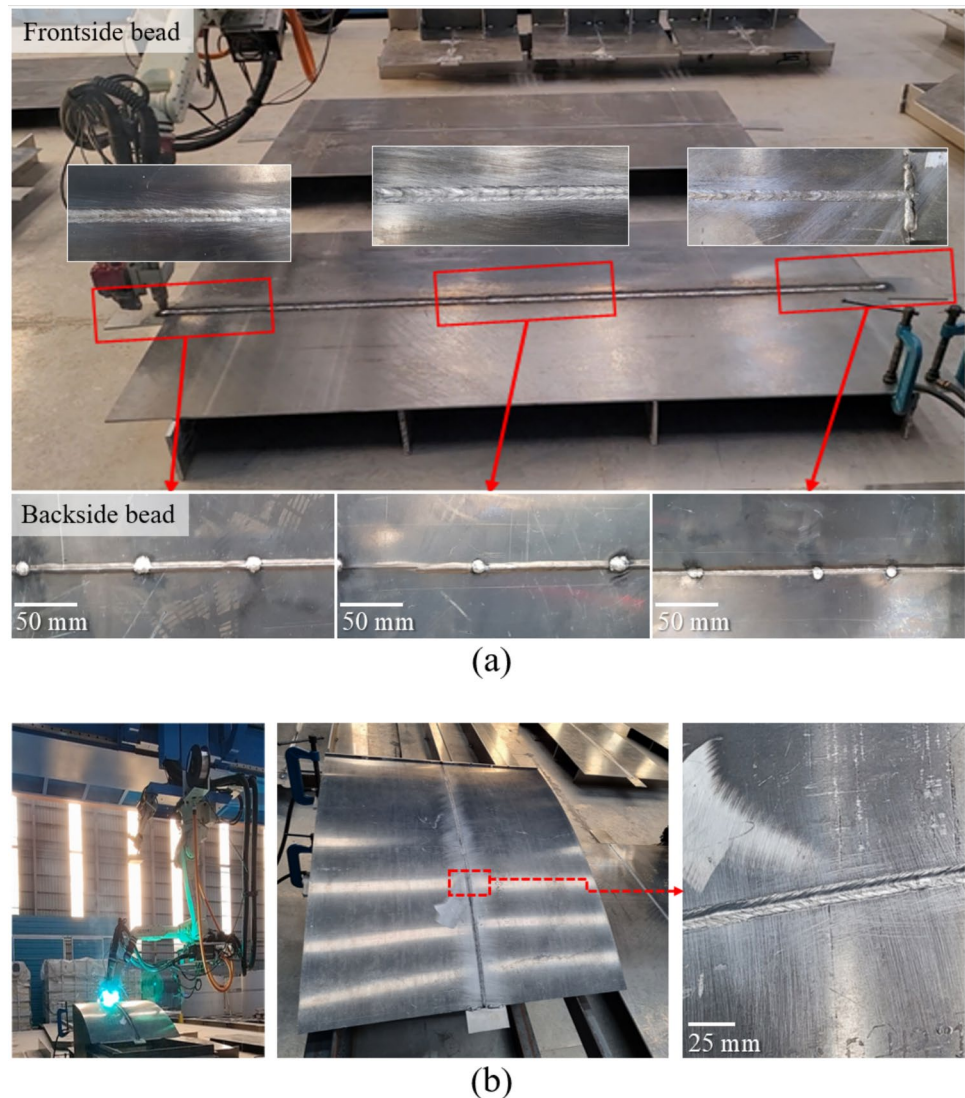
The discrepancies in displacement measurements, as detected by the GUI designed to control the laser vision sensor, ranged from 0.04 to 0.22 mm, all falling within the threshold of 0.3 mm. During actual tracking of weld seams, deviations exceeding 2 mm were rare owing to the real-time correction of deviations from the weld seam. Consequently, the configuration of the weld seam tracking system was deemed suitable for weld seam tracking applications. This precise measurement validated the accurate calibration

between the image and 3D coordinates of the laser vision sensor, as well as the coordinate transformation between the welding robot and vision sensor.

5.3 Welding experiments

Welding experiments were conducted on a butt joint specimen (workpiece 1) and a curved surface specimen (workpiece 2) to validate the welding performance of the integrated weld seam tracking and automated welding system. The details of each specimen and corresponding welding conditions are listed in Table 9. The joint welding experiment was designed to assess the lateral tracking performance of the weld seam tracking system, whereas the curved surface welding experiment aimed to evaluate the height tracking performance, ensuring a consistent distance between the welding torch and workpiece. Guides were fixed to both

Fig. 16 Experimental setup for measuring position errors between welding robot and laser vision sensor based on displacement. The position measurement using the vision sensor was performed through the GUI for laser vision sensor control



ends of each specimen to compensate for the offset distance between the vision sensor and welding torch, preserving an effective weld length for evaluation.

The welding results obtained using the weld seam tracking system are shown in Fig. 16. The visual inspection of the butt joint weld in Fig. 16a validated proper back bead formation, indicating sufficient penetration and strong joint integrity. The front bead appeared uniform and smooth, devoid of any observable defects, such as undercut, porosity, or irregularities. Similarly, in the case of curved surface welding, the front bead consistently exhibited uniform quality, with successful maintenance of weld seam tracking from the starting point to the endpoint.

In general industrial practice, manual welding is performed at speeds around 30 cm/min. The proposed system was operated at 90 cm/min while maintaining uniform bead formation and stable tracking performance throughout the weld. The welds shown in Fig. 17 were produced under the experimental conditions listed in Tables 10 and 11. To verify the internal weld quality under these conditions, a cross-sectional analysis was conducted at three representative locations, namely, start (S), middle (M), and finish (F), as illustrated in Fig. 17. The specimens were welded using highly reflective aluminum to evaluate the system robustness to specular reflection noise. Full penetration was achieved at all positions with no signs of incomplete fusion or undercut. The weld centerlines remained accurately aligned with the joint geometry, indicating stable seam tracking without lateral offset.

In both the no-gap (one-pass) and 1.5 mm gap (two-pass) conditions defined in Table 11, uniformly distributed porosity ranging from 0.2 to 0.5 mm was observed. According to ISO standards 6520–1:2012 and 5817:2014, such porosity is classified as uniformly distributed and satisfies the acceptance criteria for quality level C when considering the weld thickness ($s = 9$ mm) and base metal thickness ($a = 6$ mm) [29, 30]. This quality level is generally accepted in industrial applications such as shipbuilding and structure fabrication.

The abovementioned results confirm that the proposed system can deliver robust performance, maintaining weld seam alignment and consistent quality even at high operational speeds under various groove conditions and material reflectivity environments.

While the overall weld quality was consistent, minor keypoint detection errors were occasionally observed during seam tracking, particularly near groove corners with intense specular reflections. In such cases, the noise filtering algorithm suppressed valid laser stripe data, leading to the temporary loss of left or right corner detection. Additionally, slight deviations in keypoint position were noted and attributed to annotation bias introduced during the manual

labeling of training data. Although these problems did not considerably affect the actual weld path or integrity, as validated through the cross-sectional analysis, they indicate areas for future improvement. To further increase the system robustness, potential solutions include incorporating geometric symmetry constraints into model training, applying subpixel-level refinement techniques such as offset regression, and adopting consensus-based annotation strategies to improve the consistency of ground-truth labels.

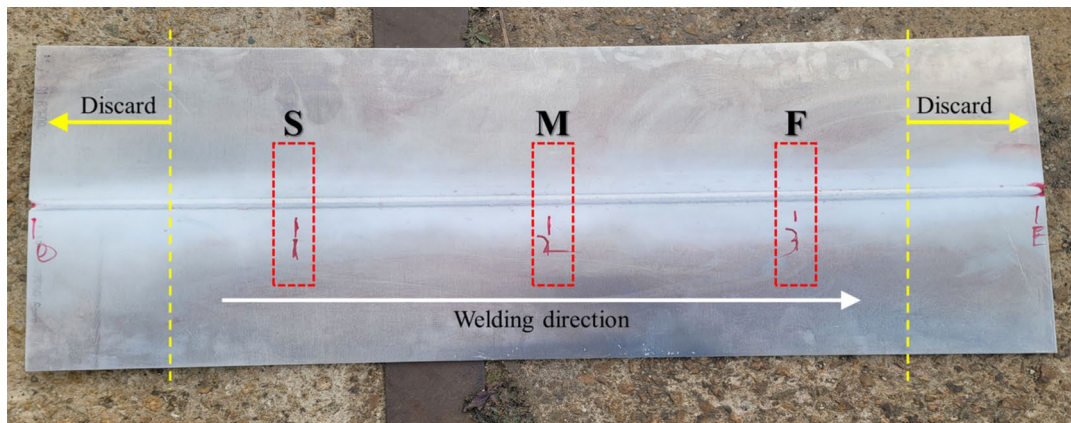
6 Conclusion

This study developed and validated an automated weld seam tracking system integrated with a multi-axis gantry-based welding robot. By leveraging a YOLOv8-pose deep learning model, the system achieved a high detection accuracy of approximately 0.128 mm, even in challenging environments characterized by fume and spatter. The model was optimized for pose-based learning of groove feature points, enabling reliable tracking performance under diverse noise conditions. These advancements underscore the potential of deep learning algorithms in overcoming limitations associated with traditional welding processes.

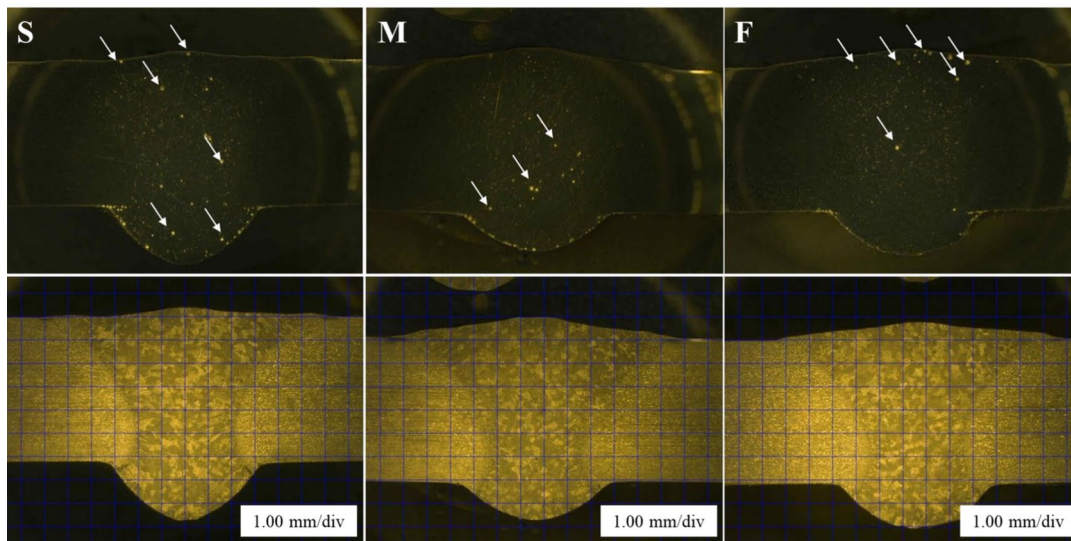
The automated system demonstrated high productivity and consistent weld quality compared with manual welding. Precise weld seam tracking was maintained throughout the process, minimizing defects, such as undercuts and incomplete penetration. Furthermore, the noise filtering algorithm contributed to the system robustness, ensuring reliable operation in real-world welding scenarios.

However, the optimal operation of the laser vision sensor necessitated the attachment of guides to the ends of the workpieces to prevent any shortening of the effective welding range caused by the offset distance between the vision sensor and welding torch. This requirement introduced an additional step into the welding process, which may limit the system applicability in certain scenarios. Future research should focus on addressing this limitation to enhance the practicality of the system and expand its adoption in industrial applications.

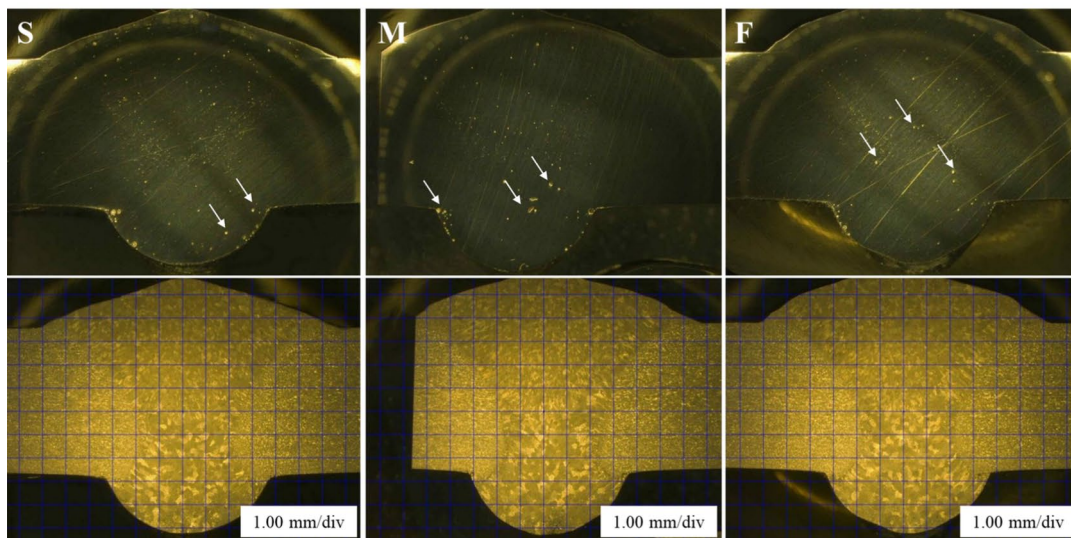
In addition, error analysis during the experiments revealed that certain detection failures were caused by severe specular reflection near groove corners, leading to the suppression of valid corner pixels by the noise-filtering algorithm. Minor position deviations were also attributed to annotation bias in the manually labeled ground truth. To address these issues, future work may incorporate strategies such as subpixel-level keypoint refinement and consensus-based annotation to reduce training error accumulation and improve the detection consistency.



(a)



(b)



(c)

Fig. 17 Cross-sectional views of welded aluminum specimens at start (S), middle (M), and finish (F) positions. **b** Single-pass weld with no gap and **c** two-pass weld with 1.5 mm gap. All sections exhibit full penetration and no visible defects such as undercut or incomplete fusion. Minor porosity (0.2–0.5 mm) is uniformly distributed and meets quality level C of standard ISO 5817

The findings of this study demonstrate the ability of the proposed system to enhance productivity, ensure uniform weld quality, and provide a reliable solution for automated welding in shipbuilding and other industrial fields. Although this study focused on shipbuilding, the proposed laser-vision-based weld seam tracking system, combined with deep learning, can be extended to other industrial applications requiring precise and robust weld tracking, such as automotive manufacturing, pipeline welding, and heavy machinery production.

To achieve broader practical applicability, several technical considerations remain to be addressed. Various challenges should be overcome for the broader application of automated welding in diverse industrial settings.

Table 10 Experimental conditions and welding parameters for macrostructure cross-sectional analysis

Parameter	Value
Workpiece	AL5083 1000 × 2000 × 6 mm 60°
Groove root	1 mm
Seam type	Butt
Process	GMAW
Weld wire	ER5356 (Alcotec)
Wire diameter	1.2 mm
Shielding gas	75% Ar + 25% He
Gas flow rate	25 l/min
Welding speed	60 cm/min

Table 11 Welding parameters for different groove gap and pass conditions

Workpiece	Parameter	Value
No gap	One pass	Current (A)
		180
		Voltage (V)
1.5 mm gap	One pass	20.0
		Wire feeding rate (m/min)
		7.0
	One pass	Current (A)
		192
		Voltage (V)
		18.8
	Two passes	Wire feeding rate (m/min)
		7.5
		Current (A)
		172
		Voltage (V)
		19.6
		Wire feeding rate (m/min)
		6.5

For example, handling complex geometries with curved or stepped surfaces, ensuring model generalization across materials with varying reflectivity, and managing sensor placement constraints in confined spaces impose practical limitations. To address these issues, the proposed YOLOv8-pose deep learning approach offers several promising directions. First, domain adaptation or transfer learning may improve generalization to different surface conditions and geometries, with domain adaptation likely enhancing the model performance in new environments that differ from the original training conditions. Second, enhancing spatial invariance through architectural adjustments or expanding the training set to include varied workpiece scenarios may help maintain detection stability. Finally, to balance precision with real-time performance, lightweight versions of the YOLOv8-pose model and multitask inference architectures are worthy of exploration in future work.

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Declarations

Competing interests The authors declare no competing interests.

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